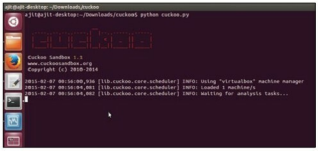

Figure 5: Cuckoo sandbox being started in the host machine

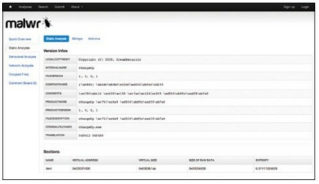

Figure 6: Malwr.com: Analysis results

Figure 7: Malwr.com: A few more analysis results

communication between host and guest. The guests should be configured with static IP addresses. (It is to be noted that DHCP is not supported.)

- Cuckoo uses an agent to establish communication between host and guest. The *agent.py* file needs to be copied into the guest.

The re-imaging tools like Fog, Clonezilla and Deepfreeze can be used for storing system images. Once the image of the guest machine is successfully taken, it shall be used by the Cuckoo sandbox. The official documentation clearly illustrates the installation process in detail.

File submission

Once the configuration is successfully done, Cuckoo (see Figure 5) can be started using the following command:

```
$ python cuckoo.py
```

The malware file can be submitted using the *submit.py* utility:

```
$ ./utils/submit.py /path/to/binary
```

Instead of this binary file, a URL could also be submitted for analysis.

Malware analysis results

After the successful completion of the analysis, the results are stored in the storage/analyses directory. The results of the analysis are given for various parameters as shown below:

- *analysis.log:* This provides information regarding the creation of processes, files and errors, if any, raised when the malware was in execution.
- *dump.pcap:* This is the network dump gathered by tcpdump.
- *memory.dmp:* This is where the complete memory dump is provided.
- The *Files* directory provides details on the files that the malware interacted with.
- The *Logs* directory provides all the log information.
- The *Reports* directory gives a list of all the reports prepared by Cuckoo.
- *Shots* lists all the screenshots made when the malware was in execution.

Malwr.com submission

If you simply require to analyse a few files and do not want to configure Cuckoo in your system, the public Web interface provided in malwr.com (as shown in Figures 6 and 7) can be used. Once the file is submitted to malwr.com, the results are provided in a clearly formatted manner through the Web interface.

Cuckoo provides an effective and efficient malware analysis mechanism. The power of Cuckoo can be understood from the testimonials provided by popular organisations like Virustotal, Threat Stream, In2Sec, etc, which use Cuckoo. Another crucial advantage of using Cuckoo is the ability to extend and customise its open source code to suit an organisation's individual requirements. Hence, Cuckoo is undoubtedly a must-have tool for security experts and malware researchers. END

References

[1] *http://docs.cuckoosandbox.org/en/latest/*
[2] *https://malwr.com/*
[3] *http://www.cuckoosandbox.org/*

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